

SIMATS ENGINEERING
ASSESSMENT TOOL 1

COURSE NAME: Deep Learning

COURSE CODE: MLA04

COURSE OUTCOME COVERED

CO1: Learning Algorithms, Maximum Likelihood Estimation (MLE), Building Machine Learning Algorithms, Neural Networks, Artificial Neurons, Perceptron, Multilayer Perceptron (MLP), Forward Propagation, Backpropagation Algorithm, Gradient Descent, Stochastic Gradient Descent (SGD), Mini-Batch Gradient Descent, Curse of Dimensionality. (BL3)

QUESTIONS: 5

Duration: 1 Hour
Total Marks:50

Annexure A

Analytical Problem Solving

1. A coin is tossed 10 times and the outcomes are:
H H T H H H T H H T
Using Maximum Likelihood Estimation, estimate the probability of getting Heads (p).
2. Out of 50 students, 42 passed an exam.
Assuming a Bernoulli distribution, find the MLE of the probability that a student passes.
3. A factory produces light bulbs. In a sample of 100 bulbs, 8 are defective.
Find the MLE for the probability that a bulb is defective.
4. A biased die is rolled 60 times. The number 6 appears 18 times.
Estimate the probability of rolling a 6 using MLE.
5. Given:
Input (x) = 4
Weight (w) = 2
Actual Output = 10
Prediction:
 $y = w \times x$
 1. Find the predicted output.
 2. Calculate the error.
 3. If the gradient is 4 and learning rate is 0.01, find the updated weight.

Answer:

1. Coin Toss – Estimate Probability of Heads (MLE)

Given:

Number of tosses = 10

Outcomes:

H H T H H H T H H T

Number of Heads = 7

Number of Tails = 3

Formula:

$$\hat{p} = \frac{\text{Number of Heads}}{\text{Total Tosses}}$$

Calculation:

$$\hat{p} = \frac{7}{10} = 0.7$$

Answer:

Estimated probability of getting Heads (MLE):

$$\boxed{p = 0.7}$$

2. Students Passing Exam – Bernoulli Distribution (MLE)

Given:

Total students = 50

Students passed = 42

Formula:

$$\hat{p} = \frac{\text{Passed}}{\text{Total Students}}$$

Calculation:

$$\hat{p} = \frac{42}{50} = 0.84$$

Answer:

MLE estimate of passing probability:

$$\boxed{p = 0.84}$$

3. Defective Light Bulbs (MLE)

Given:

Sample size = 100

Defective bulbs = 8

Formula:

$$\hat{p} = \frac{\text{Defective Bulbs}}{\text{Total Bulbs}}$$

Calculation:

$$\hat{p} = \frac{8}{100} = 0.08$$

Answer:

MLE estimate of defect probability:

$$\boxed{p = 0.08}$$

4. Biased Die – Probability of Rolling 6 (MLE)

Given:

Total rolls = 60

Number of sixes = 18

Formula:

$$\hat{p} = \frac{\text{Number of Sixes}}{\text{Total Rolls}}$$

Calculation:

$$\hat{p} = \frac{18}{60} = 0.30$$

Answer:

MLE estimate of rolling a 6:

$$\boxed{p = 0.30}$$

5. Gradient Descent Weight Update

Given:

- Input $x = 4$
- Weight $w = 2$
- Actual Output = 10
- Prediction formula:

$$y = w \times x$$

- Gradient = 4
- Learning rate = 0.01

(1) Predicted Output

$$\begin{aligned} y &= w \times x \\ y &= 2 \times 4 = 8 \end{aligned}$$

Answer:

$$\boxed{8}$$

(2) Error

$$\begin{aligned} \text{Error} &= \text{Actual Output} - \text{Predicted Output} \\ &= 10 - 8 = 2 \end{aligned}$$

Answer:

$$\boxed{2}$$

(3) Updated Weight

Gradient Descent formula:

$$w_{\text{new}} = w - \eta \times \text{gradient}$$

where

- $w = 2$
- Learning rate (η) = 0.01
- Gradient = 4

Calculation:

$$\begin{aligned}
 w_{\text{new}} &= 2 - (0.01 \times 4) \\
 &= 2 - 0.04 \\
 &= 1.96
 \end{aligned}$$

Answer:

$w_{\text{new}} = 1.96$

Final Answers

Question Answer

1 $p = 0.70$

2 $p = 0.84$

3 $p = 0.08$

4 $p = 0.30$

5.1 Predicted Output = 8

5.2 Error = 2

5.3 Updated Weight = 1.96

Code of Conduct:

I Ch.Akhil(192525250) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Ch. Akhil

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary (5 Marks)	Level 3 – Proficient (4 Marks)	Level 2 – Developing (2–3 Marks)	Level 1 – Beginning (0–1 Mark)
Problem Understanding	20%	Clearly interprets problem, identifies all parameters and requirements accurately	Understands problem with minor gaps	Partial understanding; misses key elements	Misinterprets or unable to understand problem
Method Selection	20%	Selects most appropriate method/formula with strong justification	Selects correct method with minor errors	Inappropriate or partially correct method	Unable to select method
Solution Process	25%	Logical, systematic steps with correct approach throughout	Mostly correct steps with minor mistakes	Disorganized steps; multiple errors	No clear process
Accuracy of Calculation	20%	All calculations accurate; correct final answer	Minor calculation errors	Frequent errors; incorrect final answer	No valid calculations
Interpretation of Results	15%	Clearly interprets results with units and engineering relevance	Basic interpretation with minor omissions	Limited or unclear interpretation	No interpretation